

5.2. Pick and Place Challenge

This challenge evaluates manipulation capabilities through cleaning and organizing the kitchen and preparing a simple breakfast.

Main goals:

1. Tidy up all objects on the dining table:
 - 1.1. Place dirty tableware and cutlery inside the dishwasher.
 - 1.2. Place designated trash items in the trash bin.
 - 1.3. Store other objects in the cabinet, grouping them with similar items.
2. Set up breakfast on a clean area of the dining table, including a bowl, spoon, cereal, and milk.

Optional goals:

1. Pick up trash from the floor.
2. Open and close the dishwasher door.
3. Pull and push the dishwasher rack.
4. Place a dishwasher tab inside the dishwasher.
5. Pour milk and cereal into the breakfast bowl.

Focus

Object perception, manipulation in narrow spaces, and task planning.

Setup

- **Locations:**

- **Start Location:** The robot waits outside the arena and enters when the door is opened.
- **Test location:** The test primarily takes place in the kitchen.

- **People:**

- No people are involved unless the robot explicitly requests assistance.

- **Furniture:**

- **Dishwasher:** Located near the dining table. Closed by default; the robot may request help to open or close doors or racks.
- **Dining table:** A surface in the kitchen with objects to clean up.
- **Extra Surface:** A fixed position surface with common objects (worldwide available objects). (see Figure 3.2).
- **Cabinet:** The cabinet contains objects arranged by category or similarity on different shelves.
- **Trash bin:** A trash bin is located in the kitchen.

- **Objects:**

- **Dining table setting:** Six objects arranged on the dining table in a typical post-meal setting, possibly stacked:
 - * *Cutlery:* One piece (fork, knife, or spoon).

- * *Tableware*: One mug/cup and one plate.
- * *Trash*: One trash item. One object category will be treated as trash for this task.
- * *Other objects*: Two known objects not belonging to the above categories.
- **Extra Surface setting**: Two objects from the common objects set (see Figure 3.2), and nothing else, are arranged on the extra surface.
- **Breakfast items**: The breakfast items are: a bowl, a spoon, milk and cereal. The object distribution is as follows:
 - * *Bowl and Spoon*: On top a designated surface in the kitchen.
 - * *Milk and Cereal*: Inside the cabinet, next to their respective categories. By default, the milk is closed and the cornflakes are open.
- **Cabinet objects**: A doorless cabinet. Each side of the shelves contains objects arranged in groups, either by category or likeliness.
- **Dishwasher tab**: The tab can be found on top of a designated surface and should be placed inside the dishwasher slot.
- **Floor Object**: One trash item will be placed near the trash bin.

Procedure

1. **Test start**: The robot moves to the kitchen when the arena door is open.
2. **Dining Table Clean Up**: The robot tidies the dining table by placing cutlery and tableware into the dishwasher, trash into the trash bin, and all other objects into the cabinet. Objects should be grouped by category or similarity.
3. **Serve Breakfast**: The robot sets the breakfast by placing the bowl, spoon, cereal, and milk on the dining table in a standard meal arrangement. Items should be placed with sufficient free space around them for comfortable use.
4. **Extra Surface Clean Up**: A fixed surface in the arena will hold two objects from the common objects set. These objects are placed in an easy configuration. The robot must place them into the cabinet, grouping by category or similarity, as done for the dining table clean up.
5. **Sequence**: The robot may determine the order and method for performing all pick-and-place tasks. No predefined sequence is required, and the robot may execute tasks in any order it deems optimal. *Note: Certain actions, such as pouring or closing the dishwasher, must occur after prerequisite steps (e.g., placing an item) to be scored.*

Additional Rules and Remarks

1. **First Pick Bonus**: To encourage manipulation, the robot receives an additional bonus for successfully picking the first object during the test. This bonus is awarded only once.
2. **Designated Location**: All objects designated location correspond to the appropriate furniture, either on top of or inside.
3. **Dining table Location**: The Dining table may be moved by the referee. It can be expected to be within 50 cm of its default location.
4. **Extra Surface**: The extra surface is not to be moved from its default location. It can be expected to be within 5 cm of its default location.
5. **Dining table Decoration**: The Dining table may have additional decoration, such as a flower, placed on top. The Decorations are placed at the start of the setup days.

6. **Safe placing:** Objects must be placed with care, namely the robot should place rather than throw or drop objects.
7. **Dishwasher door:** The dishwasher door is closed by default. The robot may ask for help to open or close the door or racks at any time during the task. If the robot fails to open/close the door/rack, it must clearly state this and request the referee to open/close the door/rack.
8. **Dishwasher:** The dishwasher door is closed by default. The robot may request assistance to open or close the door or internal racks at any time. If the robot fails to operate the door or racks autonomously, it must explicitly notify the referee and request help. Points for closing are only awarded after at least one item has been placed inside.
9. **Correct dishwasher item placement:** Items must be correctly positioned in the rack, as a human would place them and cleanable by the machine.
10. **Incorrect cabinet category categorization:** Objects must be grouped with similar items. Misplaced items incur score reductions. Objects that do not semantically belong to any of the categories represented on the shelves should put in an empty part of the shelf.
11. **Breakfast placement:** The table must be set in a typical setting for a meal. The spoon must be placed next to the bowl, and the cereal and milk must be placed next to each other.
12. **Breakfast area cleanliness:** The area immediately surrounding the breakfast items on the dining table must be kept clear of any other objects. Items too close at (5cm) or cluttering the space, will result in a score penalty for the breakfast placement.
13. **Pouring:** By default, the milk is closed and the cornflakes are open. If the robot needs help to open the milk container for pouring it must clearly ask for help. The robot must pour a significant amount of milk and cereal into the bowl. Pouring a couple of drops of milk or bits of cereal is not enough.
14. **Trash:** One object category will be treated as trash for this task. Announced during *Setup Days*.
15. **Human Assistance:** Scores are reduced if the robot receives help, such as pointing to objects, handing objects to the robot, repositioning items or changes to the environment. Assistance with opening the milk container or moving parts of the dishwasher does not incur a penalty. If a robot requests human assistance for environment changes (such as chairs, decorations, ...), the score reduction is applied per item. For example, if the robot requests chairs to be moved for easier access to the dining table, a score reduction is applied per chair. For decorations, the score reduction is applied per group of nearby items moved (if three candles are touching or close to each other is considered one item, however if there is a candle on each side of the dining table and robot requests a move on both, it is considered two items).
16. **Communicating Perception:** The robot must clearly indicate its perception to the referee. Pointing, attempting to pick objects, or visualizing one object at a time is sufficient. If visualization is utilized, the surrounding scene must remain visible and the robot needs to announce and confirm the referee perceived the visualization.

OC Instructions

During the *Setup Days*:

- Provide official cutlery and tableware.
- Provide official objects.
- Designate a trash category.
- Announce the dining table, extra surface and cabinet used for the test.
- Announce locations of dishwasher tab, bowl, and spoon. (in the kitchen)

Referee Instructions

The referee needs to:

- Place dining table objects (1 cutlery, 1 plate, 1 mug/cup, 1 trash, 2 other objects).
- Place extra surface objects (2 common objects).
- Place one trash on the floor (near trash bin).
- Arrange cabinet objects by category or similarity.
- Place the bowl and spoon and dishwasher tab.
- Place the milk and cereal inside the cabinet next to their respective category.

Scoresheet

The maximum time for this test is **7:00** minutes.

Action	Score
Navigate to the table	15
Correctly recognize an object	12×10
Perceive objects on a shelf and indicate the correct placement	2×30
<i>Picking</i>	
Picking up an object for transportation	12×50
First Pick Bonus	+100
From the floor	+30
Cutlery	2×+50
Plate	+100
Dishwasher tab	+100
Common object from auxiliary table	2×−20
<i>Placing</i>	
Place an object in its designated location	12×40
Correctly in the dishwasher	3×+70
Next to similar objects in the cabinet	2×+20
Area around breakfast items is not cleaned	4×−30
In the dishwasher tab slot inside the dishwasher	+160
Common object from auxiliary table	2×−20
<i>Extra Rewards</i>	
Pull or push the dishwasher rack	2×100
Open or close the dishwasher door without assistance	2×200
Open milk container without assistance	400
Pour cereal or milk into the bowl without assistance	2×200
<i>Penalties</i>	
Objects thrown or dropped while placing	12×−40
Breakfast not served in a typical meal setting	−50
Objects dropped on the floor	12×−40
Spilling cereal and milk while pouring	2×−100
Human assistance: object repositioned by a person	12×−30
Human assistance: handover	24×−100
Human assistance: environment changes (per item)	−40
Human assistance: opening milk container	−0
Human assistance: moving dishwasher door or rack	−0
<i>Special Penalties & Bonuses</i>	
Not attending (see sec. 3.8.1)	−500
Using alternative start signal (see sec. 3.6.8)	−100
Outstanding performance (see sec. 3.8.3)	351
Total Score (excluding special penalties & standard bonuses)	3515